

Kudi

An AI money agent for everyone

Intelligent money you can talk to, in your own language

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1. Executive summary

Kudi is a conversational AI agent that lets anyone manage money by talking or texting in plain language, including English, Nigerian Pidgin and Igbo. Instead of navigating a banking app, the user simply says what they want (“check my balance”, “create a card for Netflix”, “send 5k to my brother”), and the AI carries it out through the BMONI platform, always confirming before any money moves.

In one line: We are building a talk-to-your-money agent for Nigerians who find banking apps confusing, so that using powerful financial infrastructure no longer requires English fluency, app literacy or a smartphone-native mindset. Success means a live demo where an unscripted spoken request produces a real BMONI action on screen.

2. Problem statement

Who has this problem. A significant share of Nigerian adults have money to manage but are underserved by formal digital finance: low-literacy users, older users, and busy informal traders. They are not blocked by a lack of funds; they are blocked by the interface.

What the problem is. Banking and fintech apps are dense, English-first, and navigation-heavy. They assume the user is comfortable reading English, operating a multi-screen app, and interpreting financial jargon. For a large part of the population that assumption does not hold, so the app itself becomes the barrier.

Why it is painful. People avoid digital finance entirely, keep cash under stress, or depend on someone else to operate the app for them, which is slow, risky and disempowering. Meanwhile the infrastructure to serve them already exists: BMONI provides multi-currency Naira and USD wallets, instant virtual and physical cards, global transfers, currency conversion and stablecoin-backed USD savings, all through licensed partners. The missing layer is an interface simple enough that anyone can reach that infrastructure without learning an app.

The core insight. If the interface to money were a conversation in the user’s own language, the literacy and navigation barrier would largely disappear. Kudi makes the conversation the entire product.

3. Target users and personas

Primary persona — Mama Ngozi, 52, market trader, Lagos. Owns a basic smartphone. Speaks Pidgin and Igbo. Avoids banking apps because they are confusing and English-heavy. Wants to save her daily takings and pay suppliers without stress or dependence on her children. Kudi must serve her in Igbo and Pidgin, which is why Igbo is a core demo language.

Secondary persona — Emeka, 24, gig worker. Comfortable with technology but time-poor. Wants to create cards for subscriptions, move money, and save without opening five screens. He proves Kudi is faster, not just simpler.

Jobs to be done

- When I finish selling for the day, I want to move my takings into savings by just saying so, so that I save consistently without opening an app.
- When I need to pay a supplier or family member, I want to send money by speaking, so that I do not fear tapping the wrong thing.
- When I am unsure, I want the assistant to read my balance and confirm actions aloud, so that I trust what is happening to my money.

4. Strategic context and theme alignment

Why now. Two things are true at once for the first time: the financial infrastructure is mature and API-accessible (BMONI), and conversational AI is now capable enough to interpret intent across Nigerian languages and execute structured actions reliably. Kudi sits exactly at that intersection.

Mapping to the theme, “Intelligent Money for Everyone”

Judging signal	How Kudi delivers
Intelligent (AI)	An LLM agent interprets intent in English, Pidgin or Igbo and executes financial actions via tool calling
Money	Every core action is a real financial operation routed through BMONI
For everyone (inclusion)	Voice and text, local languages, and no app-navigation literacy required
Meaningful BMONI use	Balance, card creation and transfer run on BMONI; conversion and savings extend it
Working demo	A live web app where a spoken request produces a real balance read, a real card, and a real confirmed transfer

Channel strategy (stated openly). Our narrative targets low-literacy, basic-phone users, and the web app we demo is the surface that best showcases the AI and the BMONI integration to judges. The real go-to-market channel for this user is WhatsApp first, then USSD. We name this deliberately rather than pretend the web app is the finished product, and WhatsApp is the top item on our post-hackathon roadmap.

5. Goals and non-goals

Goals (for the hackathon)

- Demonstrate a working AI agent that performs real BMONI actions from natural language.
- Support English, Nigerian Pidgin and Igbo across the demo.
- Support both text and voice input, and spoken output at least in English and Pidgin.
- Make the BMONI API a central, meaningful part of the product, not decorative.
- Tell a clear inclusion story that maps directly to the theme, including an honest statement of the real delivery channel.

Non-goals (out of scope, and why)

- **Production-grade security hardening and full compliance sign-off.** Sandbox only; not shippable to real customers in 48 hours.
- **Full coverage of every BMONI endpoint.** We target a focused set to keep the demo reliable.
- **Native mobile apps.** A web app is the fastest reliable demo surface.
- **A live recurring savings engine.** Savings records the rule and performs the first save immediately; scheduling is a roadmap item.
- **Real KYC onboarding beyond sandbox requirements, and the WhatsApp/USSD channels.** Strategically the real channels, but out of scope for this build.

6. Solution overview

A web app with a single conversational surface. The user speaks or types. An AI agent (Claude with tool calling) interprets the request, confirms any sensitive action, calls the relevant BMONI endpoint through a secure server layer, and replies in the user's language by text, and by voice where a voice is available.

Design principle: the conversation is the product. No dashboards, no menus. Everything happens through dialogue, supported by lightweight visual confirmations: a card rendering on screen, a balance chip, a transfer-success state.

7. Scope

This section is the single source of truth for feature priority.

Must have (P0 — required for the demo to count as a success)

- Conversational text input and AI reply.
- Voice input in English, Pidgin and Igbo (via Whisper).
- Spoken reply in at least English and Pidgin; text reply guaranteed in all three languages.
- Multi-language understanding: English, Pidgin and Igbo.
- Check balance (BMONI read).
- Create a virtual card (BMONI write, low risk), with the card rendered on screen.
- Send money / transfer (BMONI write, high risk) as the guaranteed money-movement action.
- Explicit confirmation step before any action that moves value.

Should have (P1 — build only if P0 is solid)

- Savings rule (records the rule and performs the first save now; no recurrence engine).
- Currency conversion (NGN to USD and back), with confirmation.
- A balance chip that visually updates after a transfer.

Could have (P2 — stretch, may be framed as prototype)

- WhatsApp channel via Twilio or Meta sandbox — strategically the most valuable P2 because it matches the real target user.
- Scam-check tool: paste a suspicious alert, the AI flags it and offers to freeze the card.
- Spending summary (“how did I spend this week”).

Won't have (this hackathon)

- Full onboarding, production auth, offline mode, USSD, recurring-savings scheduler.

8. Key user flows

Flow A — Check balance (P0, read)

1. User: “How much I get?”
2. Agent calls the balance endpoint.
3. Agent replies in the user's language by text and (English/Pidgin) voice, and shows a balance chip.

Flow B — Create a card (P0, write, low risk)

4. User: “Create card make I use am for subscription.”
5. Agent confirms currency and label.
6. Agent calls the card-creation endpoint.
7. The card renders on screen and is read aloud.

Flow C — Send money (P0, write, high risk, requires confirmation)

8. User: “Send 5k give my brother.”
9. Agent normalizes the amount (5k to 5,000) and resolves the recipient from the demo beneficiary list.
10. Agent asks for explicit confirmation, restating amount and recipient.
11. On “yes”, agent calls the transfer endpoint.
12. Agent confirms success and shows the updated balance chip.

Flow D — Save money (P1)

13. User: “Save 2k for me.”
14. Agent confirms, records the savings rule, and performs the first save immediately.
15. Agent states clearly that recurrence is not active in this build.

Flow E — Convert currency (P1)

16. User: “Change 10k to dollar.”

17. Agent normalizes the amount, quotes the rate, and asks for explicit confirmation.
18. On “yes”, agent calls the conversion endpoint and confirms the new balances.

9. Functional requirements

- **FR1:** The system shall accept both text and voice input from the user.
- **FR2:** The system shall detect the user’s language (English, Pidgin or Igbo) and respond in that same language by text.
- **FR3:** The AI agent shall map natural language to a defined set of tools with structured arguments.
- **FR4:** The system shall call the corresponding BMONI endpoint for each tool and handle both success and error responses.
- **FR5:** The system shall require explicit user confirmation before any action that moves or converts value, restating amount and recipient or target currency.
- **FR6:** The system shall render a visual confirmation for balance (chip) and card creation (card render).
- **FR7:** The system shall speak responses aloud in English and Pidgin; for Igbo it shall always return a text reply and speak aloud only if a suitable voice is available, otherwise degrade silently to text.
- **FR8:** The system shall gracefully handle unsupported or out-of-domain requests with a short, helpful reply and no tool call.
- **FR9:** The savings tool shall record a rule and perform the first save immediately; it shall not imply an active recurring schedule.
- **FR10:** The system shall resolve named recipients (“my brother”) from a predefined demo beneficiary list; unknown recipients trigger a clarifying question, never a guess.
- **FR11:** The system shall normalize Nigerian shorthand amounts (for example “2k” to 2,000) before confirmation, and shall never assume an amount it was not given.

10. AI agent design

Approach. A single system prompt defines the agent as a careful, friendly Nigerian money assistant that never moves or converts value without explicit confirmation, speaks the user’s language, uses a tool for every financial action, asks a short clarifying question when unsure, and politely refuses anything outside money help.

Tools

Tool	Purpose	Priority	Risk	BMONI capability
get_balance	Read wallet balances	P0	Low, no confirm	Wallet balance (NGN / USD)
create_card	Issue a virtual card	P0	Low, no confirm	Virtual card issuance
send_money	Transfer to a recipient	P0	High, confirm required	Transfer
set_savings	Record a rule + first save	P1	Low, no confirm	Savings (USD-backed)

convert_currency	NGN to USD and back	P1	Medium, confirm required	Currency conversion
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Safety rules in the prompt

- Always restate and confirm amount and recipient (or target currency) before send_money or convert_currency.
- Never guess an amount; if the amount or recipient is unclear, ask one short question.
- Never expose raw card numbers beyond what the on-screen render intentionally shows.
- Refuse politely and stay in domain for anything outside money help.

11. Technical architecture

Frontend. Next.js and React, a single chat screen. Whisper is the primary speech-to-text engine for all three languages, because the browser Web Speech API does not reliably support Nigerian Pidgin or Igbo; the browser API is kept only as an English-only fallback. Voice output uses browser Speech Synthesis or a cloud TTS for English and Pidgin (a Nigerian-English voice renders Pidgin intelligibly). For Igbo, the app returns text and attempts speech only if a suitable voice exists, otherwise degrades to text without error.

Agent layer. Claude with tool calling. A server route receives the message, runs the agent loop, executes tool calls, and returns the reply. Conversation state is passed on each turn.

Integration layer. A thin server module wraps the BMONI API: authentication, the capability endpoints, and error mapping to friendly language. All BMONI calls happen server-side. Keys never touch the browser.

Data and hosting. Minimal: a demo session store for conversation state, the demo beneficiary list, and any recorded savings rule. Hosted on Vercel, with all sensitive keys in server environment variables only.

Speech stack

Direction	English	Pidgin	Igbo
Input (STT)	Whisper (primary)	Whisper	Whisper
Output (TTS)	Guaranteed	Guaranteed (NG voice)	Text guaranteed; speech best-effort

12. Data, privacy and compliance

- No storage of card PANs or secrets in the browser; card details are shown transiently and not persisted client-side.
- Only the minimum data needed for the demo is stored.
- Aligned with NDPR principles: collect little, explain use, allow deletion.
- Uses the BMONI sandbox and test identities only. No real customer data and no live funds.

13. Success metrics

The demo is a success if, live and without crashing, it completes:

- **Flows A, B and C** — balance, card (rendered on screen), and transfer with explicit confirmation — against the BMONI sandbox. Flow C is included specifically so the demo proves the money-movement action, not a softer substitute.
- At least two languages shown live including voice (English or Pidgin spoken; Igbo at least by text).
- At least one real BMONI action visibly happening on screen (a card appears and/or a balance updates).
- A clear one-line articulation of inclusion impact, including the honest statement of the real delivery channel.

Bonus (not required): Flow D (savings) and/or Flow E (conversion) shown as extensions.

14. Build plan (one to two day sprint)

Phase 0 — setup (first 2 hours)

- Register, obtain the BMONI API key, read the docs, list the actual endpoints and auth method.
- Confirm which capabilities are supported in the sandbox; balance and card are the confidence anchors.
- Scaffold the Next.js app and repo; stand up the Whisper path early since it is on the critical path.

Phase 1 — spine (next 4 hours)

- Build the BMONI server wrapper for `get_balance` and `create_card`.
- Wire the Claude agent loop with those two tools.
- Basic chat UI (text only) plus the card render and balance chip.

Phase 2 — the wow (next 4 hours)

- Add voice input (Whisper) and voice output (English/Pidgin).
- Add language handling; test Pidgin and Igbo end to end.
- Add `send_money` with the confirmation gate (prioritize over any P1).
- Only after Flow C works: attempt P1 (`set_savings`, then `convert_currency`).

Phase 3 — polish and demo (final hours)

- Error handling and graceful fallbacks, including STT and TTS degradation paths.
- Rehearse the 90-second demo end to end at least three times.
- Record a backup video of a clean run; prepare the pitch and submission blurb.

15. Team

Team Fintech Disruptors, organised around three functional roles:

- **Product and AI:** system prompt, tool schemas, agent loop, demo narrative and pitch.
- **Backend and integration:** BMONI wrapper, endpoints, auth, error handling, deployment.
- **Frontend and voice:** chat UI, Whisper speech-to-text, text-to-speech, card and balance visuals.

16. Dependencies and risks

Dependencies

- BMONI sandbox access, API key and documentation (resolved in Phase 0).
- Whisper access for local-language speech-to-text.
- A cloud TTS with a Nigerian-English voice for natural Pidgin output (browser TTS is an acceptable fallback).
- Claude API access with tool calling.

Risks and mitigations

Risk	Impact	Mitigation
BMONI sandbox lacks a needed endpoint	High	Confirm in Phase 0. Keep balance and card live. Clearly label anything unsupported in the demo.
Whisper struggles on heavy accents or Igbo	Medium	Text input is a first-class path; tune prompts; never block the demo on speech.
No Igbo voice available for output	Low	FR7 degrades Igbo to text; inclusion still holds via Igbo text plus English/Pidgin speech.
Money movement risky in a live demo	Medium	Sandbox and test identities only; always confirm before send; predefined beneficiary list removes ambiguity.
Scope creep into P1/P2 before P0 is solid	High	Section 7 governs; do not start savings or conversion until Flow C works.
Live network issues during the demo	Medium	Pre-recorded backup video of a successful run.

17. Demo script (90 seconds)

The script leads with the three guaranteed flows so nothing on stage depends on optional features.

19. **Balance (Pidgin):** “How much I get?” Kudi replies in Pidgin by voice and shows the balance chip.
20. **Card (Pidgin):** “Create one card make I use am for subscription.” A real BMONI virtual card renders on screen and is read aloud.
21. **Transfer (the centerpiece):** “Send 5k give my brother.” Kudi restates “Send 5,000 Naira to Chidi, your brother, yes or no?” On “yes” it executes the BMONI transfer and the balance chip updates. This one moment shows intelligence, real money movement and safety together.
22. **Close line:** “Money anyone can use, just by talking, in their own language, and it always asks before it moves your money. That is intelligent money for everyone. And on WhatsApp, it reaches the people who need it most.”

18. Assumptions and open questions

Assumptions

- The BMONI sandbox exposes, at minimum, wallet balance, virtual card issuance and transfer.

- Conversion and savings are assumed available and are treated as P1 because they are unconfirmed until day zero.

Open questions (resolved on day zero)

- Final product name (Kudi is a working title).
- Whether a suitable Igbo voice is available for spoken output, or whether Igbo stays text-only.
- Whether the sandbox provides a test recipient for the transfer demo, or whether we seed our own.

19. Post-hackathon roadmap

23. WhatsApp channel (highest priority) — the channel that actually reaches the target user.
24. USSD fallback for non-smartphone users.
25. A real recurring savings engine so saving rules genuinely recur.
26. Broader language coverage (Yoruba, Hausa) once the three-language core is proven.
27. Production security, auth and KYC for a shippable product.

20. Appendix: BMONI integration checklist (day zero)

- Authentication method and sandbox base URL.
- Endpoint for wallet and balance retrieval.
- Endpoint and parameters for virtual card creation.
- Endpoint for transfers and how recipients are represented.
- Currency conversion support and how rates are returned.
- Savings support and whether it is USD/stablecoin-backed.
- Rate limits and test credentials.